

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)			Indicative content: <u>Absorbers:</u> Explain how absorbers can be used to distinguish between the three radiations – describe set up. Diagram or a clear explanation. Sheet of paper stops alpha. Few mm of aluminium will stop alpha and beta. Gamma penetrates both the paper and aluminium. Reference to background radiation. <u>Magnetic field / Electric field:</u> Explain how a magnetic / electric field can be used to distinguish between the three radiations – describe set up. Diagram and clear explanation. Deflection of alpha particles much smaller than for beta, charge positive. Deflection of beta particles opposite to that of alpha, as charge-negative. No deflection of gamma. Correct direction of alpha or beta in magnetic / electric field. 5-6 marks Absorbers and magnetic / electric field both covered comprehensively. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks Either absorbers or magnetic / electric field covered comprehensively or limited account of both areas. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i>	4	2		6		6